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The Verb-Movement Parameter in Second Language Acquisition

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This article investigates effects of instruction on parameter resetting in second language acquisition, where the first and second language (French and English, respectively) differ as to the settings they adopt for verb movement (Pollock, 1989). The question addressed is whether instruction on one of a cluster of properties associated with lack of verb movement in English (i.e., question formation) generalizes to another property associated with lack of movement (i.e., adverb placement). A total of 138 francophone learners of English as a second language (ages 10-12) were exposed to two different conditions, being instructed either on English adverb placement or on question formation. Subjects were tested on their knowledge of adverb placement using three different tasks. They were tested prior to instruction, immediately after instruction, and again after a delay of 5 weeks. Results show clear differences between the groups; only the subjects instructed on adverbs came to know the restrictions on adverb placement imposed by lack of verb movement in English. The implications of these results for parameter resetting in L2 acquisition are discussed.

1. INTRODUCTION

Government Binding (GB) Theory has a dual aim: to characterize the native speaker's knowledge of language, or linguistic competence, and to explain how the acquisition of such competence is possible. In GB Theory (as well as in earlier versions of generative grammar), it is argued that much of our linguistic competence stems from innate knowledge, which takes the form of a Universal Grammar (UG). Linguists motivate UG by pointing to the end result of first language (L1) acquisition, namely the adult grammar in

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all its complexity, arguing that there is no way this could be acquired without prior knowledge of some kind, given the kind of input that children are exposed to.

Assuming that abstract principles and parameters of UG constrain L1 acquisition, what is their relevance to second language (L2) acquisition? If, as I believe, a major task of L2 acquisition theory is to describe and explain the acquisition of L2 competence, then naturally the question arises as to whether L2 learners still have access to the abstract principles and parameters of UG, that is, the extent to which their competence resembles that of the L1 learner. Do L2 learners, for example, have knowledge of abstract structural relationships like c-command? What is the effect on the learner when a parameter requires different settings in the L1 and the L2? Do parameters show the same clustering effects in L2 acquisition as they do in L1?

Such questions can only be explored in the context of a theory of linguistic competence. Current linguistic theory offers a highly detailed account of what linguistic competence consists of, as well as an indication of how that competence is acquired. Even if it turns out that L2 learners do not attain the complex and abstract knowledge that would be expected if UG is still available, this is something that can only be determined by investigating universal principles and parameters as they are isolated by linguists. An L2 acquisition theory can take as a working hypothesis that L2 learners do (or do not) still have access to principles and parameters of Universal Grammar, in order to establish the exact nature of L2 competence and the mechanisms involved in L2 acquisition. In this article, I assume certain recent theoretical proposals for a parameter of verb movement, and investigate the operation of this parameter in L2 acquisition.

2. SOME PROPERTIES OF FRENCH AND ENGLISH

The experimental study to be described here is concerned with certain parameterized differences between French and English, the question of whether these differences lead to an acquisition problem for the L2 learner, and the kind of evidence that a learner might require to arrive at the correct properties of the L2. Pollock (1989) and Chomsky (1989), following earlier work by Emonds (1978, 1985), proposed a parameter that accounts for a number of differences between these two languages, including adverb placement, negative placement, and question formation, described as follows.

French and English contrast in certain respects as far as adverb placement is concerned. In French, an adverb may appear between the verb and its direct object (SVAO), whereas in English it may not. French (1a) is grammatical, whereas its English equivalent, (1b), is ungrammatical.[1]

(1) a. Jean embrasse souvent Marie.
b. *John kisses often Mary.

In French, an adverb may not appear between the subject and the verb (SAV), whereas this is possible in English, as shown in (2).

(2) a. *Jean souvent embrasse Marie.
b. John often kisses Mary.

These languages share the possibility of allowing an adverb to occur after an auxiliary verb, as in (3).

(3) a. Jean a souvent embrassé Marie.
b. John has often kissed Mary.

The two languages also behave differently as far as negative placement is concerned, as can be seen in (4). In French, the negative *pas* is found after the main verb, as in (4a), whereas English *not* cannot occur in this position, as in (4b). Instead, it must be placed after an auxiliary verb, such as *do* in (4c).

(4) a. Marie n'aime pas Jean.
b. *Mary likes not John.
c. Mary does not like John.

French and English differ with respect to question formation as well. (5a) shows that the main verb in French can invert with the (pronominal) subject to form a question. In contrast, English does not allow subject-verb inversion but rather requires subject-auxiliary inversion, as shown in (5b) and (5c).

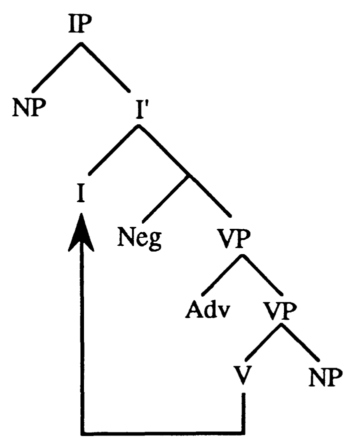
(5) a. Aime-t-elle Jean?
b. *Likes she John?
c. Does she like John?

[1]In all the examples, the English (b) sentence is a word-for-word translation of the French (a) sentence; consequently, glosses are not provided.

3. THE VERB-MOVEMENT PARAMETER AND ADVERB PLACEMENT

It has been argued by Pollock (1989) and Chomsky (1989) that these apparently different properties can all be traced to one parametric difference between the two languages, namely whether or not the language allows verb raising. French has verb movement: All finite verbs must raise to Infl, in contrast to English verbs, which may not raise (with the exception of *have* and *be*). The underlying structure of English and French sentences containing adverbs is the same, in that adverbs are optionally base-generated adjoined to the verb phrase (VP).[2] This structure together with verb movement is shown in (6).[3]

(6)



This analysis accounts for the similarities and differences between French and English as follows. As far as the adverb positions are concerned, the finite verb must raise in French, giving the order SVAO, as in (1a). Failure

[2]Pollock specifically proposed this position for frequency adverbs (*often, seldom,* etc.) and did not discuss manner adverbs. Emonds (1976) proposed that manner adverbs are generated at the end of the VP, with a transformational rule moving them to the front. Jackendoff (1972) suggested that there are several possible base-generated positions for manner adverbs, including the front of the VP. Thus, either by base-generation or by movement, VP-initial is a possible adverb position for various classes of adverbs.

[3]I have oversimplified the issues in two ways here. Pollock argued that the category Infl must be split into two, Tense and Agr, each heading a projection, with the verb moving through both. The parameter he proposed involves "opacity" versus "transparency" of the category Agr, not simply ± verb movement. These distinctions allowed him to account for many other phenomena in the two languages, such as differences in the behavior of finite and nonfinite clauses. Some implications for L2 acquisition of his more articulated structure are pursued in White (to appear).

to raise results in the ungrammatical SVA, as in (2a). Verb raising is prohibited in English; thus SVAO cannot occur, as in (1b), and SAV is found, as in (2b). The only verbs that raise in English are *have* and *be*. They raise to Infl over the adverb, just as they do in French, giving the order SauxAVO, as in (3a) and (3b). As for negatives, the obligatory raising of the finite verb in French explains why negative *pas* is postverbal in (4a). Questions are formed by "subject-verb inversion," as in (5a), because the verb subsequently raises from Infl to Comp, that is, to the left of the subject. The lack of verb raising in English accounts for the impossibility of postverbal negation, as in (4b), and explains why main verbs do not invert in questions, as in (5b).

4. LEARNABILITY AND PARAMETER SETTING

4.1 L1 Acquisition

Where a parameter involves a cluster of properties, the assumption is that these properties do not have to be learned individually by the L1 learner. Rather, evidence from one aspect of the cluster should be sufficient to trigger all consequences of the parameter setting (Chomsky, 1981). In the case of the Verb-Movement Parameter, the French child could get evidence from questions, negatives, or adverb placement that verb movement is possible in French. Suppose, for example, that the French child uses negatives in the input (particularly the position of *pas* with respect to the finite main verb) as evidence of verb movement, then one would not expect the child to have to learn separately that verb raising is also possible in questions or past adverbs.[4] Similarly, in the case of English, evidence of *do* support in negatives, say, should be sufficient to indicate that main verbs do not move, with the associated consequences for the grammar of English. Either value of the parameter, then, should be learnable on the basis of readily available positive input. Therefore, I assume that this parameter is "open" in L1 acquisition, that is, it does not have a default setting.

4.2 L2 Acquisition

If UG operates in L2 acquisition exactly as it does in L1, then one might expect that parameter setting in the second language would be no different than parameter setting in the first. In other words, UG would "start over," with open parameter values, or unmarked settings in some cases. If so, the

[4]See Pierce (1989) for evidence that very young children acquiring French as their L1 do indeed assume verb movement.

learner of French would take evidence such as the position of *pas* to indicate the possibility of verb raising, whereas the learner of English would take *do* support as evidence to the contrary.

However, a number of researchers have argued that the L2 learner does not approach the L2 with parameters of UG unset. Rather, the parameter setting operating in the first language has various effects on the way the learner approaches the L2 input. Experimental evidence suggests that, in some cases, the L1 parameter setting is incorrectly assumed to be appropriate for the L2, though not necessarily permanently so (e.g., Hilles, 1986; Phinney, 1987; White, 1985). Differences in parametric values in the L1 and L2 may cause other effects, such as delay in acquiring the appropriate properties of the L2 (Flynn, 1987). Where a parameter involves a cluster of properties, this cluster may not work together in the L2 as it does in the L1 (Liceras, 1989; White, 1985).[5]

If the learner initially approaches the L2 data from the perspective of the L1 parameter setting, then the French learner of English has to discover that English does not allow verb movement. Properties of English such as the use of *do* support in negatives and questions indicate that the main verb does not move into Infl; such sentences are presumably common in the input and could provide the learner with positive evidence that verb raising does not apply in the L2, as pointed out by Schwartz (1987). Furthermore, the existence of SAV order in English, as in (2a), also constitutes positive evidence that verbs do not raise. On the other hand, there are data that show that French learners of English have problems with adverb placement, persistently producing and accepting sentences like (1b), even when they are at advanced levels (Sheen, 1980; White, 1989a). Such data suggest that the L2 positive input is not sufficient to indicate that verb movement is prohibited past an adverb.

In the rest of this article, I investigate the francophone learner's knowledge of English adverb placement, before and after instruction. Learners are exposed to two different kinds of instruction. Subjects in one condition are specifically instructed on only one aspect of the Verb-Movement Parameter, namely its effects on adverb placement, receiving positive evidence as to the possibility of SAV order in English as well as negative evidence as to the impossibility of SVAO order. Subjects in the other condition receive instruction on a related aspect of the parameter, namely question formation, and no specific instruction on adverb placement. This allows one to see whether instruction on one of a cluster of properties associated with lack of verb movement in English (i.e., question formation) generalizes to another property associated with lack of movement (i.e., adverb placement).

[5]For more detailed discussion of these issues, see White (1989b).

5. THE EXPERIMENT

5.1. Subjects

Subjects were child native speakers of French, taking part in intensive English as a second language (ESL) programs in the Province of Quebec, Canada. Five classes participated (ranging in size from 25 to 30 students per class), two at the Grade 5 level (average age 11 at time of this study) and three at Grade 6 (average age 12). These children had little exposure to English prior to entering the program and are effectively at the beginner level. The program involves 5 months of intensive ESL instruction, and almost no activities are conducted in French during this time period; however, it is not an immersion program, as no other subjects are taught in English. The emphasis is on communicative language teaching, and there is normally little use of form-focused instruction or error correction.

The study involved two experimental conditions: one Grade 5 and two Grade 6 classes (82 children in all) were assigned to be taught certain aspects of English adverb placement; one Grade 5 and one Grade 6 class (56 children in all) were instructed in question formation. In addition, there was a control group of 26 Grade 4 and 5 monolingual native speakers of English.

5.2 Timetable and Research Design

After approximately 3 months in the program, to allow students to become sufficiently proficient in English to participate in the study, all classes were pretested on adverb placement. Up to this point, none of the classes had had any instruction on adverbs. Immediately after this pretesting, the teachers of the adverb group introduced teaching materials and activities on adverb placement, which they taught for the 2 subsequent weeks. The question group was taught question formation during the same time period. All classes were then retested on adverb placement (first posttest). A second posttest was administered at the end of the intensive program, approximately five weeks after the first.[6] The research design is summarized in Table 1.

5.3 Teaching Materials and Input

Teaching materials were specially prepared for this study and were developed in a tightly prescribed manner to ensure that the teachers would teach

[6]This study is also reported and discussed, from a different perspective, in White (1991), which includes additional results from a long-term follow-up. In addition, subjects were tested on question formation, as reported in White, Spada, Lightbown, and Ranta (1991).

TABLE 1
Adverb Placement Study: Research Design

	Adverb Group (n = 82)	*Question Group (n = 56)*	*Control Group (n = 26)*
	Grades 5 and 6	*Grades 5 and 6*	*Grades 4 and 5*
Pretesting (Day 1)	Pretesting on adverbs	Pretesting on adverbs	Testing on adverbs
Teaching (2 weeks)	Teaching on adverbs	Teaching on questions	
1st posttest (Day 15)	Posttesting on adverbs	Posttesting on adverbs	
2nd posttest (Day 50)	Posttesting on adverbs	Posttesting on adverbs	

the same activities in the same order, in the same manner, and for the same period of time. This was accomplished by giving each teacher a package of instructional materials (either on adverb placement or question formation) with specific guidelines for their implementation. Teachers instructing the adverb classes were also taped while teaching, and all teachers filled out a questionnaire, which later provided additional information concerning any changes or revisions they made while working through the materials (these turned out to be minimal).

Teachers in the adverb condition spent 5 hr in the first week conducting intensive work on adverb placement, with the second week devoted to 2 hr of follow-up activities. Teachers in the question condition spent a corresponding amount of time on question formation. Teaching concentrated on two kinds of English adverb: adverbs of frequency (*sometimes, often,* etc.) and adverbs on manner (*quickly, slowly,* etc.). Emphasis was on adverb meanings (using contextualized activities) and form (specifically, positions in SVO structures). Many of the teaching materials and activities were form-focused, giving explicit positive evidence that adverbs can appear in a variety of positions in the sentence (ASVO, SAVO, SVOA), as well as negative evidence indicating that they may not appear between verb and object (*SVAO).

Teachers and children were encouraged to point out and correct errors. Instruction for the question group focused on word order, with emphasis on placement of the subject and the auxiliary verb and on the circumstances in which *do* support is required. At no time did any of the teachers have access to the tests that were administered to their students.

In addition to this controlled input, there would, of course, be the normal input of classroom interaction. Judging from tapes of teachers involved in such programs, teachers frequently use questions and negatives in class. Thus, classroom input in the form of questions and negatives could provide additional positive evidence that the verb does not raise in English, and both the adverb group and the question group would have received such input. Spontaneous use of adverbs in the classroom, on the other hand, appears to be minimal.

It was important that the children not get misleading input from their teachers and that the teachers notice and correct any relevant errors. Consequently, only native speakers of English were chosen to take part in this study, because native speakers of French who are otherwise very accurate bilinguals often persist in using SVAO word order (Sheen, 1980).

5.4 Tests

Three different tasks were devised to test knowledge of English adverb placement. The following frequency and manner adverbs were tested:

often, always, sometimes, usually, quickly, slowly, quietly, carefully. The following adverb positions were tested with transitive verbs: ASVO, SAVO, SVAO, SVOA. In addition, some sentences tested adverb placement in the case of intransitive verbs followed by a prepositional phrase (PP). Although it would have been desirable to have included a spontaneous production task, it is in fact very difficult to devise a test that leads to spontaneous use of adverbs, so none was included.

All three classes in the adverb group and one class in the question group also took the English proficiency test of the Ministry of Education of Quebec (MEQ test).

5.4.1 Grammaticality Judgment Task

One of the tests was a written grammaticality judgment/correction task that took the form of a cartoon story. Subjects had to read the story and indicate any cases of incorrect word order. There were 33 sentences in the story: 16 involved adverb positions (both permissible and impermissible), 10 were other grammatical sentences, and 7 were ungrammatical distractors.

5.4.2 Preference Task

Another test was a written preference task. Subjects had to read pairs of sentences and then circle one of the responses written beneath the pair, as in (7).

(7) a. Linda always takes the metro.
  b. Linda takes always the metro.
only a is right only b is right both right both wrong don't know

Such a task has the advantage of limiting what the subject has to judge, as two sentences are presented for consideration that differ only in syntactic form. At the same time, subjects are nevertheless giving an outright judgment in the case where one sentence is preferred over the other.

There were four versions of this test, each consisting of 32 sentence pairs, of which 28 dealt with adverb positions and the rest were distractors. Two versions had different sentences in them, and each of these versions occurred in two orders. Subjects were randomly assigned to a particular version at the pretesting and subsequently took the same version in a different order.

5.4.3 Manipulation Task

For the third task, subjects were tested individually. Adverb placement was tested with sets of words on cards (one word per card) that could be used to form sentences. Subjects would be handed the first set (randomly shuffled) and asked to lay out an English sentence using all the words; each set always included an adverb. Then they were asked if they could make another sentence using the same words, and so on until they could do no more. They were then presented with the next set of cards, and the procedure was repeated. Responses were written down by an experimenter. There were two versions of this test, each testing two of the frequency and two of the manner adverbs, three of the sentences being SVO in form, and one being SVPP. Children were randomly assigned to one or the other version at each test session. Each child manipulated four different sentences at each test session, responding with anything from one to four different orders for each sentence.

The judgment and preference tasks took on average 15 min each to complete, and individual testing on the manipulation task took approximately 7 min per child.

5.5 Results

Results from all three tasks reveal clear differences between the adverb group and the question group. There proved to be no significant differences between Grades 5 and 6; consequently, results from the different grade levels are collapsed.

The difference between the mean test scores on the MEQ test of English proficiency was significant, $F(3, 106) = 10.48, p = .0001$. Post hoc Scheffé procedures ($p < .05$) show that the three classes instructed on adverbs were not significantly different from each other and that the scores of the Grade 6 question class were significantly higher than those of two of the adverb classes. Due to an administrative oversight, the children in the Grade 5 question class did not take this test, but informal assessments of their English proficiency by their teacher suggest that they were in no way unusual; that is, they would have fallen within the normal range found in these programs.

5.5.1 *The Grammaticality Judgment Task (Cartoon)*

On the grammaticality judgment task, each subject was assigned an SVAO error score, consisting of responses to any SVAO sentences that were left unchanged (a maximum of four in this test) plus any other sentences whose order was incorrectly changed to SVAO.

The test contained seven ungrammatical distractor sentences to make sure that subjects were capable of judging and correcting incorrect sentences and that they were paying attention. Subjects who altered fewer than three of the distractors have been eliminated from the analysis of this task, because they appear to have a tendency not to change sentences in general. In such a case, failing to identify an incorrect SVAO sentence would not be very revealing of their competence on adverb positions in English. This leaves 37 of the adverb group and 38 of the question group who passed the distractor criterion each time they took this test. Their SVAO scores are presented in Figure 1.

A repeated measures ANOVA shows that difference between the mean scores of the adverb and question groups is highly significant, $F(1, 73) = 89.61$, $p = .0001$, as are differences at the three test sessions $F(2, 73) = 49.43$, $p = .0001$. The interaction between groups and test sessions is also significant, $F(2, 73) = 39.13$, $p = .0001$. Post hoc Scheffé procedures ($p < .05$) show that there is no significant difference between question and adverb groups prior to instruction, and both differ significantly from the controls. At the two posttests, the performance of the group instructed on adverbs is not significantly different from the controls. There is no significant difference between the question group's scores on the three tests, suggesting no improvement over time in the absence of specific teaching on adverb placement. In contrast, the adverb group's pretest performance differs from both posttests, and the two posttests do not differ from each other, suggesting that they learn that SVAO order is prohibited in English and do not forget what they are taught.

The teaching materials did not include instruction on placement differences between frequency and manner adverbs. Manner adverbs usually sound better at the end of the VP, whereas frequency adverbs sound better in the SAV position (i.e., near Infl). Many students in the adverb group nevertheless demonstrated knowledge of this difference, suggesting that they were able to extract information from the positive input even when it

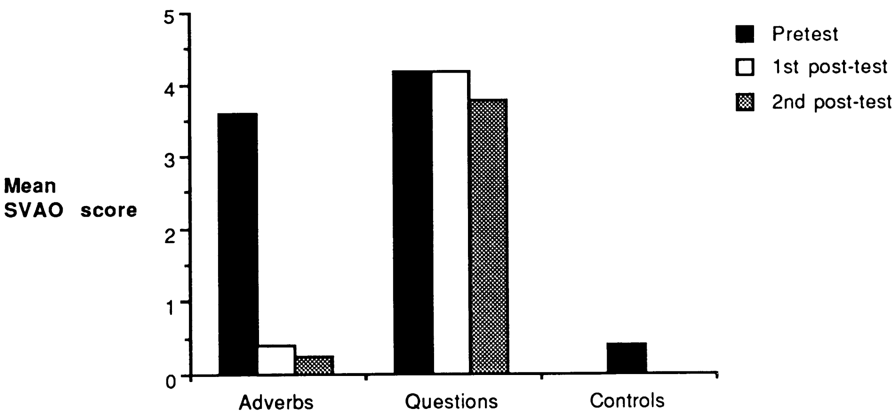


FIGURE 1 Cartoon task: SVAO error scores.

was not specifically drawn to their attention by the teacher. When correcting SVAO sentences in the cartoon task, the native-speaker control group repositioned frequency adverbs in the SAV position and manner adverbs in the SVOA position. This pattern of error correction was also adopted by two of the three classes that were taught adverbs. The children instructed on question formation in general failed to correct these errors at all.

Subjects were also assigned an SAV score on this task, calculated by adding responses to any SAV sentences that were left unchanged (maximum four) plus any other adverb sentences whose order was changed to SAV. These results are presented in Figure 2. A repeated measures ANOVA again shows significant differences between the adverb and question groups, $F(1, 73) = 25.89$, $p = .0001$, and the three test sessions, $F(2, 73) = 21.59$, $p = .0001$. The interaction between group and test is highly significant, $F(2, 73) = 16.57$, $p = .0001$. Post hoc Scheffé procedures ($p < .05$) reveal no significant difference between question and adverb groups prior to instruction, and a significant difference after instruction. It is clear that subjects in the question group do not totally reject SAV order in English, but neither do they use it as much as the native-speaker controls or those specifically instructed about adverb placement, who achieve scores not significantly different from the native speakers.

5.5.2 *Preference Task*

SVAO error scores were also calculated for the preference task; 12 sentence pairs compare an ungrammatical SVAO position with some other grammatical adverb position. The error score for an individual subject consists of any preferences for the sentence with SVAO order, together with

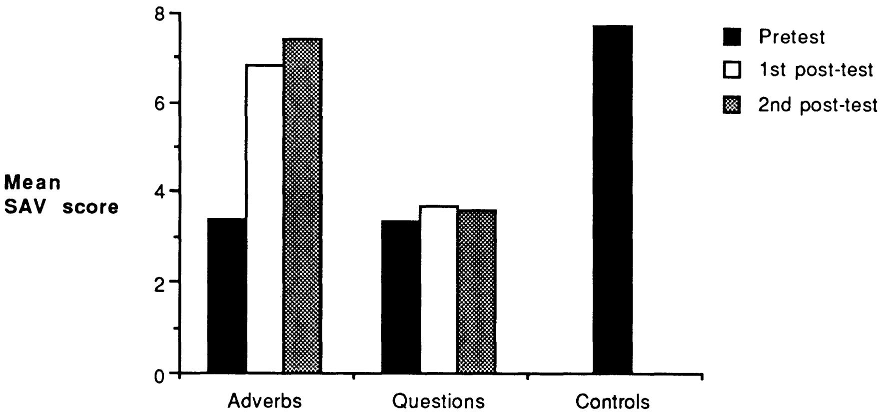


FIGURE 2 Cartoon task: SAV scores.

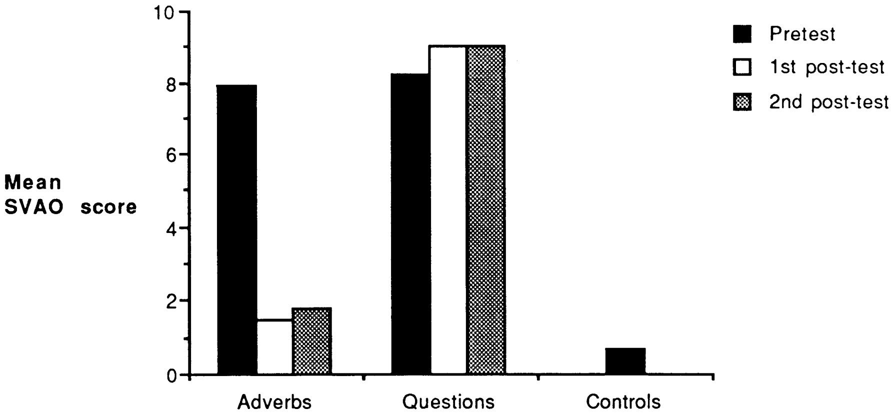


FIGURE 3 Preference task: SVAO error scores.

any answers that rate the SVAO and the other sentence to be "both right." These results are presented in Figure 3. The analyses involve 68 children in the adverb condition and 46 children in the question condition who took the preference task on all three occasions.

As in the grammaticality judgment task, a repeated measures ANOVA shows a significant effect for group, $F(1, 112) = 251.463$, $p = .0001$, for the test session, $F(2, 112) = 96.207$, $p = .0001$, as well as a significant interaction, $F(2, 112) = 99.294$, $p = .0001$. The post hoc Scheffé tests show no significant difference between question and adverb groups on the pretest, and significant differences between the two conditions on the two posttests, with the adverb group's error scores not differing significantly from the controls.

Figure 4 presents the SAV scores for this task. Sixteen of the sentences compared an adverb in the SAV position with an adverb in some other position. The SAV score for an individual subject consists of any preferences for the sentence with SAV order, as well as any answers that rate the SAV and the other sentence to be "both right."[7] The native speakers rarely chose some other adverb position in preference to SAV, a pattern also achieved by the group instructed on adverb placement. A repeated measures ANOVA shows a significant effect for group, $F(1, 112) = 23.75$, $p = .0001$, and for test session, $F(2, 112) = 143.334$, $p = .0001$, as well as a

[7]The SAV score does not necessarily reflect accuracy, because in some cases "both right" is the correct answer, whereas in other cases "only SAV" is correct, depending on the position of the adverb in the comparison sentence. There is therefore some room for error in this score, for example, if the subject rates on SAV and SVAO pair as "both right." Nevertheless, the score is included, as it reflects increasing sensitivity to the fact that SAV is a possible position in English.

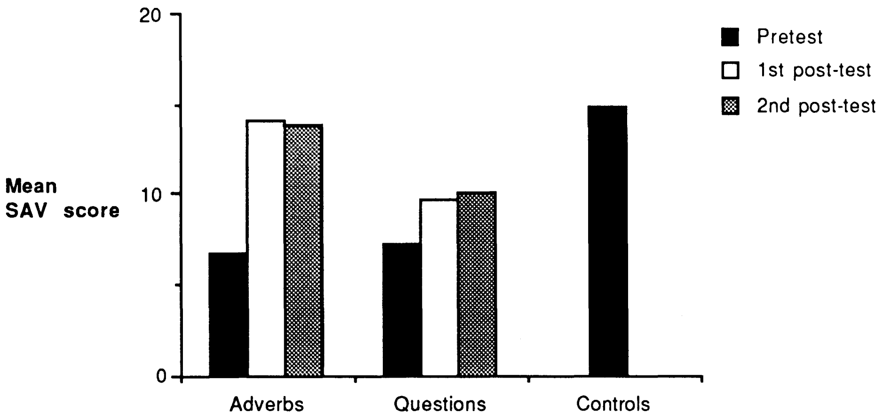


FIGURE 4 Preference task: SAV scores.

significant interaction, $F(2, 112) = 25.779, p = .0001$. As before, post hoc Scheffé procedures ($p < .05$) show that the adverb and question group means are not significantly different at pretesting and that they differ significantly at both posttests. It is clear, however, that there is an increase in acceptance of SAV order not only by the adverb group but also by the question group. The difference between the question group's mean SAV scores at the pretest and first posttest, and at the pretest and second posttest, is significant.

In order to look in more detail at changes in subjects' treatment of the grammatical SAVO and ungrammatical SVAO word orders, Tables 2 and 3 present the results from the sentence pairs that compare an SAVO sentence with SVAO, in order words, using sentence pairs similar to (7). Combining the different versions of the preference task yields eight exemplars (using the eight test adverbs) of each word-order pair for comparison. In other words, there are four pairs of sentences comparing a frequency adverb in

TABLE 2
Preference Task With Frequency Adverbs: Responses Comparing SAVO and SVAO, in Percentages

		Pretest		*1st Posttest*		*2nd Posttest*	
	Controls	*Adverbs*	*Questions*	*Adverbs*	*Questions*	*Adverbs*	*Questions*
Both right	1.92	31.65	36.11	4.58	40.74	5.59	43.27
Both wrong	1.92	7.59	6.48	2.61	8.33	2.48	1.92
Only SAVO	96.15	24.05	19.44	87.58	14.81	85.09	23.08
Only SVAO	0.00	34.18	36.11	4.58	35.19	6.21	31.73
Don't know	0.00	2.53	1.85	0.65	0.93	0.62	0.00

TABLE 3
Preference Task With Manner Adverbs: Responses Comparing SAVO and SVAO, in Percentages

		Pretest		*1st Posttest*		*2nd Posttest*	
	Controls	*Adverbs*	*Questions*	*Adverbs*	*Questions*	*Adverbs*	*Questions*
Both right	3.85	24.68	36.11	3.92	52.34	6.83	54.81
Both wrong	1.92	9.49	6.48	7.19	4.67	4.35	7.69
Only SAVO	92.31	11.39	14.81	83.66	13.08	81.37	11.54
Only SVAO	0.00	51.27	41.67	4.58	29.91	7.45	25.96
Don't know	1.92	3.16	0.93	0.65	0.00	0.00	0.00

SAVO position with one in SVAO position, and four comparing manner adverbs in these two positions.

The results from pairs involving frequency adverbs are presented in Table 2, those from manner adverbs in Table 3. Each table represents a total of 843 responses. Both tables show that subjects give predominantly wrong answers on the pretests: The most prevalent responses are that both sentences are correct or that only the SVAO order is correct. (The incidence of responses indicating only SAVO to be correct, however, suggests that this order is not rejected outright, as do responses indicating that both orders are correct.) Learners' responses contrast with the native speakers, whose overwhelming response (96.15% and 92.31%) is to say that only SAVO is correct. After the teaching period, the responses of the adverb group pattern with the controls, and this remains true at the second posttest. The children instructed on questions show little change over time in their responses to the frequency adverbs. However, their responses to manner adverbs show an increase in the response of "both right" and a corresponding decrease in the response indicating only the SVAO sentences to be correct, suggesting that they are becoming more aware that SAV is a permissible English order, although remaining unaware that SVAO is not permissible.

In order to see what kinds of generalizations subjects might make when faced with sentence types that had not been covered by their instruction, the test also included sentences with transitive verbs followed by prepositional phrases, with adverbs in the positions shown in (8).

(8) a. John slowly walks to school (SAVPP).
    b. John walks slowly to school (SVAPP).

Manner adverbs are acceptable in positions like (8b), presumably by base-generation (see footnote 2); in other words, verb raising has not taken place in such cases. However, frequency adverbs sound bad in this position (compare *John walks slowly to school* with *John walks often to school*). If subjects unconsciously know the difference between verb phrases containing a transitive or an intransitive verb, and the implications for adverb placement of this difference, they should accept sentences like (8b). On the other hand, if the effect of instruction on adverb placement is local to sentence types explicitly covered in classroom input, subjects may not be able to handle pairs like these.

Table 4 shows the distribution of responses to sentence pairs involving manner adverbs like (8). As in the case in Tables 2 and 3, Table 4 represents a total of 843 responses. The control group overwhelmingly considers both versions of the sentence to be correct (86.54%). Before instruction, the predominant response of both adverb and question groups favors SVAPP order. After instruction, the adverb group favours only the SAV order, and this is true at the first and the second posttest. It appears that these subjects may be overgeneralizing certain aspects of their instruction, incorrectly assuming that SVAO and SVAPP orders are alike in prohibiting a postverbal adverb. In contrast, a comparison of the pretest and posttest responses of the question group shows an increase in the choice of the response of "both right," suggesting that these subjects are becoming more aware of the possibility of SAV order in English.

It might appear from these results that the group not specifically instructed in adverb placement is more successful at handling adverb placement with intransitive verbs than the instructed group. However, the results from frequency adverbs show that the question group accepts SVAPP order, an order not, on the whole, acceptable to native speakers. Table 5 gives the distribution of responses to sentence pairs like (9).

(9) a. Mary often walks to school (SAVPP).
b. Mary walks often to school (SVAPP).

Table 5 shows that the native speaker control group has a strong preference for SAVPP order (75% of their responses). After instruction,

TABLE 4
Preference Task With Manner Adverbs: Responses Comparing SAVPP and SVAPP, in Percentages

		Pretest		*1st Posttest*		*2nd Posttest*	
	Controls	*Adverbs*	*Questions*	*Adverbs*	*Questions*	*Adverbs*	*Questions*
Both right	86.54	18.47	28.7	11.04	50.93	10.62	51.92
Both wrong	0.00	4.46	11.11	5.84	4.63	6.25	4.81
Only SAVPP	5.77	10.19	11.11	76.62	18.52	73.75	18.27
Only SVAPP	7.69	63.69	42.59	5.19	24.07	8.12	25.00
Don't know	0.00	3.18	6.48	1.3	1.85	1.25	0.00

TABLE 5
Preference Task With Frequency Adverbs: Responses Comparing SAVPP and SVAPP, in Percentages

		Pretest		*1st Posttest*		*2nd Posttest*	
	Controls	*Adverbs*	*Questions*	*Adverbs*	*Questions*	*Adverbs*	*Questions*
Both right	15.38	26.75	29.91	7.19	41.67	4.94	47.12
Both wrong	0.00	7.01	9.35	2.61	7.41	2.47	5.77
Only SAVPP	75.00	18.47	16.82	81.7	12.96	82.1	12.5
Only SVAPP	5.77	43.95	38.32	8.5	36.11	9.26	32.69
Don't know	3.85	3.82	5.61	0.00	1.85	1.23	1.92

the adverb group shows a similar preference. The question group, on the other hand, shows an increase in acceptance of both orders, as they did for manner adverbs (Table 4). This suggests, once again, that although exposure to question formation may be sensitizing them to the possibility of SAV order in English, they are not able to distinguish when SAV is the only acceptable order.

5.5.3 The Sentence Manipulation Task

In the sentence manipulation task, subjects were assigned an SVAX score and an SAV score, presented in Figures 5 and 6, respectively. The SVAX score represents the number of occasions on which the subject laid out a sentence with SVAO or SVAPP order, whereas the SAV score represents the number of occasions on which the subject laid out an SAV order. The maximum possible score is 4 in each case, as each subject was given four different sentences to manipulate. Seventy-two children in the adverb

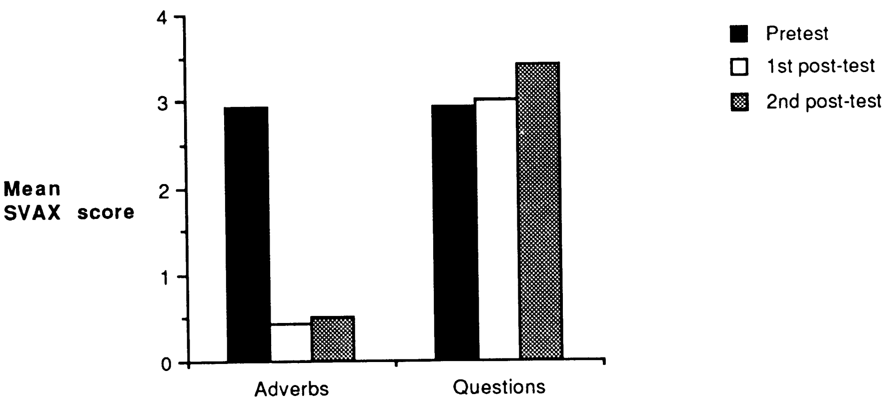


FIGURE 5 Manipulation task: SVAX scores.

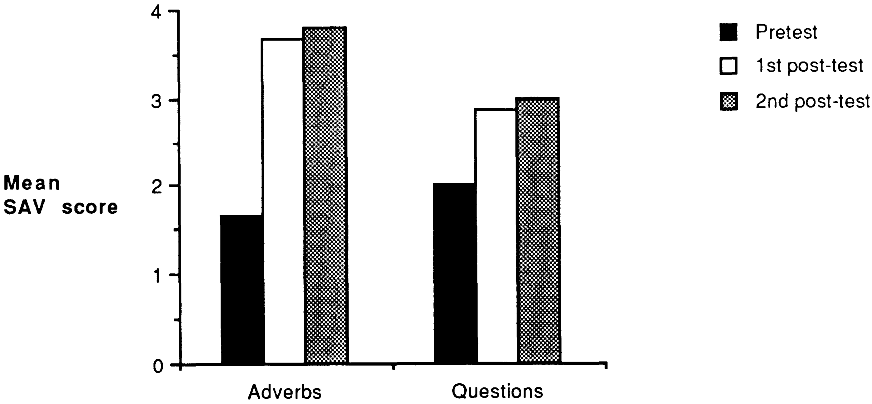


FIGURE 6 Manipulation task: SAV scores.

classes and 43 children in the question classes took this test on all three occasions. The results from this task show the same trends as those from the other tasks.

A repeated measures ANOVA shows highly significant differences between the mean SVAX scores of the two groups, $F(1, 1133) = 183.91, p = .0001$, and the three test sessions, $F(2, 113) = 84.1, p = .0001$, as well as a significant interaction between groups and test sessions, $F(2, 113) = 71.71, p = .0001$. Post hoc Scheffé procedures ($p < .05$) show that the incidence of SVAX order by the question group on the second posttest is significantly higher than on the pretest or the first posttest, suggesting that continued exposure to positive input does not lead to a realization that certain orders are ungrammatical.

The SAV scores on the manipulation task are shown in Figure 6. A repeated measures ANOVA shows highly significant differences between adverb and question groups, $F(1, 113) = 6.91, p = .01$, and the three test sessions, $F(2, 113) = 114.1, p = .0001$, as well as a significant interaction, $F(2, 113) = 13.99, p = .0001$. Post hoc Scheffé procedures ($p < .05$) show that the increase in use of this order between the pretest and the first posttest is significant for both groups.

6. DISCUSSION

In summary, pretest results from both groups on all tasks show that francophone learners of English accept and use the ungrammatical SVAO word order, suggesting that they incorrectly assume that English allows the verb to raise past an adverb, as it does in French. Second, there are clear differences between the classes instructed on adverbs and those instructed on questions. Only the adverb group learns that SVAO order is ungrammatical, and the adverb group shows a much more dramatic increase in acceptance and use of SAV order than the question group. Although the classes instructed in question formation did show an increase in SAV order on two of the three tasks, they did not show any corresponding decline in their acceptance or use of SVAO order.

In the short term, at least, it appears that instruction on one of the properties associated with lack of verb movement in English, namely, lack of verb raising in question formation, does not generalize to another property associated with lack of movement, namely conditions on adverb placement. Only the subjects instructed on adverbs came to know the restrictions on adverb placement. However, it should be noted that a long-term follow-up of one of these classes conducted one year after the original study (with no further instruction on either adverb placement or question formation) shows that most of the differences between the groups disappear (White, 1991). Subjects instructed on adverb placement revert to the behavior they showed at pretesting, accepting and producing SVAO order. These results thus suggest that explicit instruction is ineffective in the long run, regardless of which aspect of the parameter is focused on. (See White, 1991, for more details and a discussion of possible reasons for this failure.)

In certain aspects, the short-term effects of instruction appear to have been quite local. The instruction on adverb placement focused on transitive verbs, and subjects made inappropriate generalizations when faced with intransitive verbs. (However, this failure does not tell us anything specific about the Verb-Movement Parameter, because verb raising is not at issue in the intransitive sentences.) Nevertheless, there were certain respects in which subjects did make correct generalizations beyond their specific instruction. Two of the three adverb classes demonstrated knowledge of a distinction between preferred positions for manner and frequency adverbs, and the question group did increase its acceptance and use of SAV order.

Results from this study suggest that supplying positive evidence from one of a cluster of properties relating to a parameter does not have extensive effects on the L2 acquisition of other aspects of the cluster. In this case, explicit evidence on adverb placement was more effective in helping L2 learners master the fact that English disallows verb raising past an adverb than explicit evidence on question formation. Although both forms of input resulted in an increased appreciation of what is permissible in English, the adverb group showed far greater acceptance and use of SAV order than the question group, and only the adverb group showed evidence of knowing the impossibility of SVAO order, at least in the short run.

The pretest results suggest that SVAO and SAV word orders are co-occurring in the grammars of most subjects before instruction, and this remains true of the subjects who were taught questions. (In the long run, it is also true of the subjects who were taught adverb placement.) This does not appear to fit either value of the parameter, which requires raising with finite verbs in French, resulting in SVAO and *SAV, and prohibits it in English, resulting in *SVAO and SAV.[8] Only the group specifically instructed on adverb placement achieves the latter combination.

There are thus two ways in which properties associated with the Verb-Movement Parameter do not appear to cluster together in L2 acquisition. First, instruction on one aspect of the L2 parameter setting does not generalize to other aspects. Second, the interlanguage grammars of L2 learners appear to allow possibilities that are inconsistent with either value of the parameter.

There are a number of potential explanations of the failure of these L2 learners to show the expected clustering. One possibility, of course, is that parameters of UG no longer operate in L2 acquisition (Bley-Vroman, 1989; Clahsen & Muysken, 1986; Schachter, 1988). This would account both for the failure of instruction on question formation to have any effects on adverb placement and for the fact that learners allow possibilities inconsistent with either value of the parameter. However, it is generally accepted by the researchers cited that UG is only unavailable to adult learners and that it is likely to operate in child L2 acquisition. In that case, the causes of failure to cluster must lie elsewhere.

Another possibility is that UG is indeed available but that the rather conscious kind of instruction that these learners received would not provide the appropriate input to reset parameters of UG (see Schwartz, 1987). This would account for the failure of instruction in the long run but would not explain why naturalistic positive input in the classroom (in the form of questions and negatives) appeared to have no effect in helping learners to discover that English prohibits verb raising past adverbs.

Yet another possibility is that L2 learners take the positive evidence (including evidence from question formation) not as evidence against verb raising but as evidence for its optionality. That is, positive evidence such as existence of *do* support shows that the verb need not raise (hence, SAV

[8]English has another adverb position, between the subject and Infl, as in *John probably has left.* This position does not relate to the verb-raising issue; in such a case, an adverb can also appear between the auxiliary and main verb, as in *John probably has quietly left,* showing that even after the raising of *have,* there is another available adverb position. French does not have this position. This means that there are actually two different reasons why SAV is unfamiliar to French learners of English: (a) obligatory verb raising in French and (b) no position between subject and Infl. None of our sentences had a lexicalized Infl, so the SAV position is ambiguous between SAIV and SIAV. If our subjects are assuming the former, this has nothing to do with the Verb-Movement Parameter.

order is permissible), but this does not drive out the possibility that it might sometimes raise (hence, SVAO order is also found). According to Pollock (1989), verb movement in French is optional in the case of nonfinite verbs, so this possibility is not ruled out in principle. Learning that something is obligatory when you think it is optional has always been recognized as causing a learnability problem, and this may be the reason why those instructed specifically on adverb placement do better than those who get evidence from question formation. Only the former group was specifically instructed as to the impossibility of SVAO order, effectively being shown that movement is not optional but prohibited.[9]

Finally, Pollock's analysis might be incorrect. Iatridou (1990) argued that adverb-placement differences between English and French are not, in fact, a consequence of differences in verb-raising possibilities. If she is right, we are dealing with two different phenomena here, and it is not surprising that instruction on question formation has no effect on adverb placement; they are not part of the same parameter.

Advances in linguistic theory and advances in L2 acquisition theory are often inextricably intertwined. We are faced here with a familiar situation: Data from L2 learners suggest that a clustering of properties associated with a parameter does not hold; at the same time, linguistic theory suggests that the parameter may not be exactly as originally formulated (cf. the history of the Null-Subject Parameter within linguistic theory and within L2 acquisition research). It thus becomes impossible, given our current state of knowledge, to decide which of the alternatives just discussed provides the correct explanation of the phenomena reported here. The answer must await further research.

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[9]Francophone learners of English appear to assume optional "short movement" to Agr but not "long movement" to Tense. The present study was not designed to tease these positions apart, but in a related study, conducted on similar groups, evidence of long movement was never found (White, to appear).

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REFERENCES

Bley-Vroman, R. (1989). What is the logical problem of foreign language learning? In S. Gass & J. Schachter (Eds.), *Linguistic perspectives on second language acquisition* (pp. 41-68). Cambridge, England: Cambridge University Press.

Chomsky, N. (1981). *Lectures on government and binding.* Dordrecht, The Netherlands: Foris.

Chomsky, N. (1989). Some notes on economy of derivation and representation. *MIT Working Papers in Linguistics, 10,* 43-74.

Clahsen, H., & Muysken, P. (1986). The availability of Universal Grammar to adult and child learners: A study of the acquisition of German word order. *Second Language Research, 2,* 93-119.

Emonds, J. (1976). *A transformational approach to English syntax.* New York: Academic.

Emonds, J. (1978). The verbal complex V′-V in French. *Linguistic Inquiry, 9,* 151-175.

Emonds, J. (1985). *A unified theory of syntactic categories.* Dordrecht, The Netherlands: Foris.

Flynn, S. (1987). *A parameter-setting model of L2 acquisition.* Dordrecht, The Netherlands: Reidel.

Hilles, S. (1986). Interlanguage and the pro-drop parameter. *Second Language Research, 2,* 33-52.

Iatridou, S. (1990). About AgrP. *Linguistic Inquiry, 21,* 551-557.

Jackendoff, R. (1972). *Semantic interpretation in generative grammar.* Cambridge, MA: MIT Press.

Liceras, J. (1989). On some properties of the "prodrop" parameter: Looking for missing subjects in non-native Spanish. In S. Gass & J. Schachter (Eds.), *Linguistic perspectives on second language acquisition* (pp. 109-133). Cambridge, England: Cambridge University Press.

Phinney, M. (1987). The pro-drop parameter in second language acquisition. In T. Roeper & E. Williams (Eds.), *Parameter setting* (pp. 221-238). Dordrecht, The Netherlands: Reidel.

Pierce, A. (1989). *On the emergence of syntax: A crosslinguistic study.* Unpublished doctoral dissertation, Massachusetts Institute of Technology.

Pollock, J. (1989). Verb movement, Universal Grammar, and the structure of IP. *Linguistic Inquiry, 20,* 365-424.

Schachter, J. (1988). Second language acquisition and its relationship to Universal Grammar. *Applied Linguistics, 9,* 219-235.

Schwartz, B. (1987). *The modular basis of second language acquisition.* Unpublished doctoral dissertation, University of Southern California, Los Angeles.

Sheen, R. (1980). The importance of negative transfer in the speech of near-bilinguals. *IRAL, 18,* 105-119.

White, L. (1985). The pro-drop parameter in adult second language acquisition. *Language Learning, 35,* 47-62.

White, L. (1989a). The principal of adjacency in second language acquisition: Do L2 learners observe the subset principle? In S. Gass & J. Schachter (Eds.), *Linguistic perspectives on second language acquisition* (pp. 134-158). Cambridge, England: Cambridge University Press.

White, L. (1989b). *Universal grammar and second language acquisition.* Amsterdam: Benjamins.

White, L. (1991). Adverb placement in second language acquisition: Some effects of positive and negative evidence in the classroom. *Second Language Research, 7,* 133–161.

White, L. (to appear). Long and short verb movement in second language acquisition. In C. Lefebvre, J. Lumsden, & L. Travis (Eds.), *Functional category theory.*

White, L., Spada, N., Lightbown, P. M., & Ranta, L. (1991). Input enhancement and L2 question formation. *Applied Linguistics, 12.*